AI/ML PROJECT SUBMISSION

PROJECT OVERVIEW	STUDENT DETAILS
• **Project Number:** 4 • **Task Name:** Quiz and Exam System • **Difficulty:** Moderate	• **Name:** Pranav Gajanan Rane • **College:** Government Polytechnic Gondia • **Branch:** Computer Engineering • **Domain:** AI/ML

Project 3 : Quiz & Exam System

The Quiz and Examination System is an interactive Python application that presents multiple choice questions (MCQs) to a student, evaluates responses, tracks scores, and generates a detailed result report at the end. The question bank is stored using tuples (immutable, for data integrity) and the application handles timing, scoring, and grade calculation automatically.

CODE-

```
#--------------------------------------------------------------------------------
'''
Project 4 - Quiz & Examination System
Made by Pranav Rane
'''
#--------------------------------------------------------------------------------

'''Functions Definitions followed by actual system.'''
```

```
import random # Used to shuffle our quedtions each time

def load_questions(): # Questions for the quiz is stored here with the Ansers
  questions = [
    ("Which data type is immutable in Python?", "List", "Dictionary", "Tuple", "Set", "C"),
    ("What is the output of print(2 ** 3)?", "6", "8", "9", "12", "B"),
    ("Which keyword is used to define a function in Python?", "func", "define", "def", "function", "C"),
    ("What does len('hello') return?", "4", "5", "6", "error", "B"),
    ("Which of the following is a mutable data type?", "Tuple", "String", "List", "Integer", "C"),
    ("What symbol is used for single-line comments in Python?", "//", "/* */", "#", "--", "C"),
    ("What will type(3.14) return?", "int", "float", "str", "double", "B"),
    ("How do you take input from the user in Python?", "scan()", "cin()", "input()", "readline()", "C"),
    ("What does the append() method do to a list?", "Removes last element", "Adds element at beginning", "Adds element at end", "Sorts the list", "C"),
    ("Which loop is best when number of iterations is already known?", "while", "do-while", "for", "repeat", "C"),
    ("What is the index of the first element in a Python list?", "-1", "1", "0", "None", "C"),
```

```
("Which function is used to find the length of a list?", "size()", "count()", "len()", "length()", "C"),
 ]
 return questions


def display_question(qno, q): # Used to display the question with their options
 print(f"Q{qno}: {q[0]}")
 print("OPTIONS - ")
 print(f" A) {q[1]}")
 print(f" B) {q[2]}")
 print(f" C) {q[3]}")
 print(f" D) {q[4]}")


def get_answer(): # To take answer as an input from the user
 while True:
  try:
   ans = input("Your Answer: ").upper()
   if ans in ["A", "B", "C", "D"]:
    return ans
   else:
    print("Please enter A, B, C or D options only.")
  except Exception:
   print("Invalid input. Try again.")


def calculate_grade(percentage): # Showcases the percentage
```

```
if percentage >= 90:
    return "O"
  elif percentage >= 80:
    return "A+"
  elif percentage >= 70:
    return "A"
  elif percentage >= 60:
    return "B+"
  elif percentage >= 50:
    return "B"
  else:
    return "F"


def show_wrong_answers(wrong_list):  # Displays Wrong answers choosen
  if len(wrong_list) == 0:
    print("\nNo wrong answers. Great job!")
    return

  print("\n--- Wrong Answers Review ---")
  for entry in wrong_list:
    qno = entry[0]
    q = entry[1]
    your_ans = entry[2]
    options = {"A": q[1], "B": q[2], "C": q[3], "D": q[4]}
    correct = q[5]
    print(f"\Q{qno}: {q[0]}")
          print(f" Your Answer : {your_ans} - {options[your_ans]}")
           print(f" Correct Ans : {correct} - {options[correct]}")
```

```
def show_report(name, roll, score, total, wrong_list):
# Shows the final report for the quiz
 percentage = (score / total) * 100
 grade = calculate_grade(percentage)

 if percentage >= 50:
  result = "PASS"
 else:
  result = "FAIL"

 print("\n====== QUIZ RESULT ======")
 print(f"Name   : {name}")
 print(f"Roll No  : {roll}")
 print(f"Score   : {score} / {total}")
 print(f"Percentage : {percentage:.2f}%")
 print(f"Grade   : {grade}")
 print(f"Result  : {result}")
 print("=============================")

 show_wrong_answers(wrong_list)


def evaluate_quiz(name, roll, questions): # Initializes
the quiz
 score = 0
 wrong_list = []
 total = len(questions)

 for i in range(total):
  display_question(i + 1, questions[i])
  ans = get_answer()
```

```
correct = questions[i][5]

    if ans == correct:
      score += 1
      print("Correct!")
    else:
      print(f"Wrong! Correct answer was: {correct}")
      wrong_list.append((i + 1, questions[i], ans))

  show_report(name, roll, score, total, wrong_list)


def main(): # main method
  print("=====++++PYTHON QUIZ++++=====")

  name = input("Enter your name: ")

  roll = None
  while roll is None: # Essential for one time use
    try:
      roll = int(input("Enter Roll Number: "))
    except ValueError:
      print("Roll number should be an integer.")

  print(f"Student: {name} | Roll: {roll}")

  questions = load_questions()
  random.shuffle(questions)
```

```
print(f"Total Questions Available: {len(questions)}")
 print("Enter choice A, B, C or D for each question.")

 evaluate_quiz(name, roll, questions)


main()
```

Note - This code might break due to incorrect indentation while pdf making so pls use the code mentioned in .py file attached

TEST OUTPUTS -

```
=====++++PYTHON QUIZ++++=====
Enter your name: Pranav
Enter Roll Number: 15
Student: Pranav | Roll: 15
Total Questions Available: 12
Enter choice A, B, C or D for each question.
Q1: What will type(3.14) return?
OPTIONS -
  A) int
  B) float
  C) str
  D) double
Your Answer: b
Correct!
```

```
Q2: What symbol is used for single-line comments in Python?
OPTIONS -
  A) //
  B) /* */
  C) #
  D) --
Your Answer: c
Correct!
```

```
Q11: Which loop is best when number of iterations is already known?
OPTIONS -
  A) while
  B) do-while
  C) for
  D) repeat
Your Answer: d
Wrong! Correct answer was: C
```

```
====== QUIZ RESULT ======
Name       : Pranav
Roll No    : 15
Score      : 9 / 12
Percentage : 75.00%
Grade      : A
Result     : PASS
===========================
```

```
--- Wrong Answers Review ---
\Q7: Which of the following is a mutable data type?
  Your Answer  : A - Tuple
  Correct Ans  : C - List
\Q11: Which loop is best when number of iterations is already known?
  Your Answer  : D - repeat
  Correct Ans  : C - for
\Q12: How do you take input from the user in Python?
  Your Answer  : A - scan()
  Correct Ans  : C - input()
```

TEST OUTPUTS IN BROKEN CASES -

```
Q1: Which loop is best when number of iterations is already known?
OPTIONS -
  A) while
  B) do-while
  C) for
  D) repeat
Your Answer: 65
Please enter A, B, C or D options only.
Your Answer: 
```

```
====== QUIZ RESULT ======
Name        : Yash
Roll No     : 98
Score       : 0 / 12
Percentage  : 0.00%
Grade       : F
Result      : FAIL
===========================
```

CODE TESTING WAS DONE SUCCESSFULLY!

More Details about the project is mentioned in the .md file attached . This pdf was made specifically to attach the code and its outputs.

PROJECT COMPLETED